



COURSE SLIDES

Voice and Video Agents with n8n (C500)

Course Code: C500

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Tertiary Infotech Academy Training Team

Principal Trainer
Tertiary Infotech Academy Pte.
Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Practitioner in workflow automation, conversational AI and synthetic media production.

DELIVERS

Applied AI, agent automation and no-code workflow courses.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1 Your role and one communication process you would like to improve.

2 Your current experience with n8n, voice interfaces or video creation.

3 Which matters most for your use case: response speed, consistency, scale or accessibility?

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

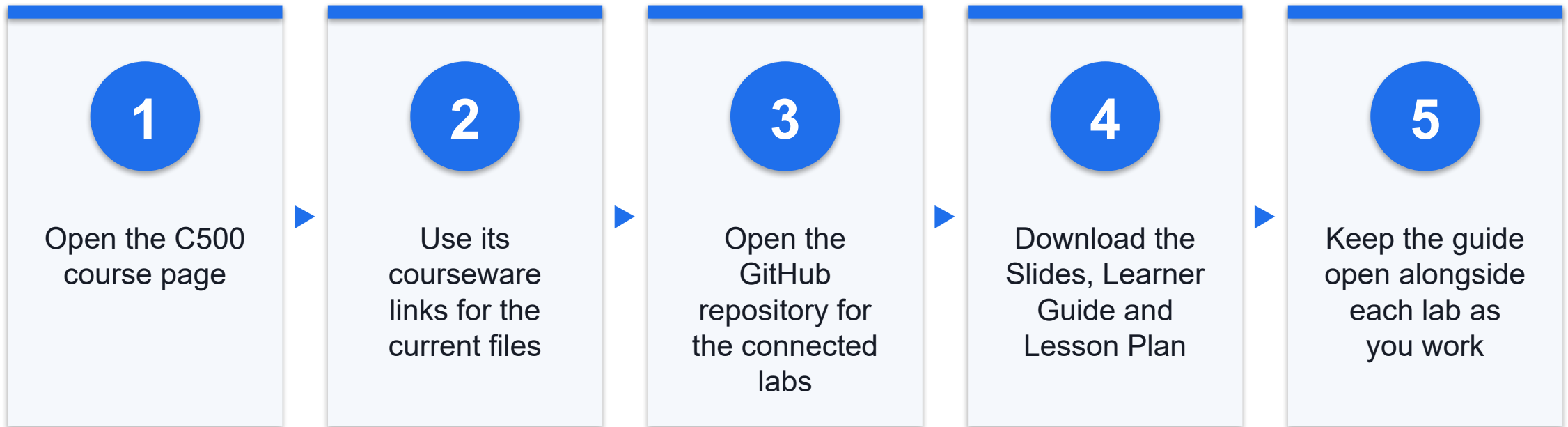
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Open Course Material



Labs: <https://github.com/tertiarycourses/C500-Voice-and-Video-Agents-with-n8n>

Lesson Plan — 1 Day, 8 hours/day

Day 1

- Day 1 — Build and govern voice and avatar-video agents with n8n
 - Topic 1: Voice Agent (Labs 1–2)
 - Topic 2: Video Agent (Labs 3–4)

Timing

- Daily timing
 - 9:00 am - 6:00 pm (1-hour lunch; tea breaks within scheduled time)
- Learning rhythm
 - Concept → demonstrate → build → verify → recap

Learning Outcomes

1

Voice Architecture

Explain the listen-think-talk architecture of a voice agent, select suitable business use cases, and design the n8n workflow and channel boundary.

2

Working Voice Agent

Build and verify an n8n voice agent that transcribes audio with Whisper, reasons with an AI Agent, speaks with ElevenLabs, and operates through browser and chat channels.

3

Avatar Video Agent

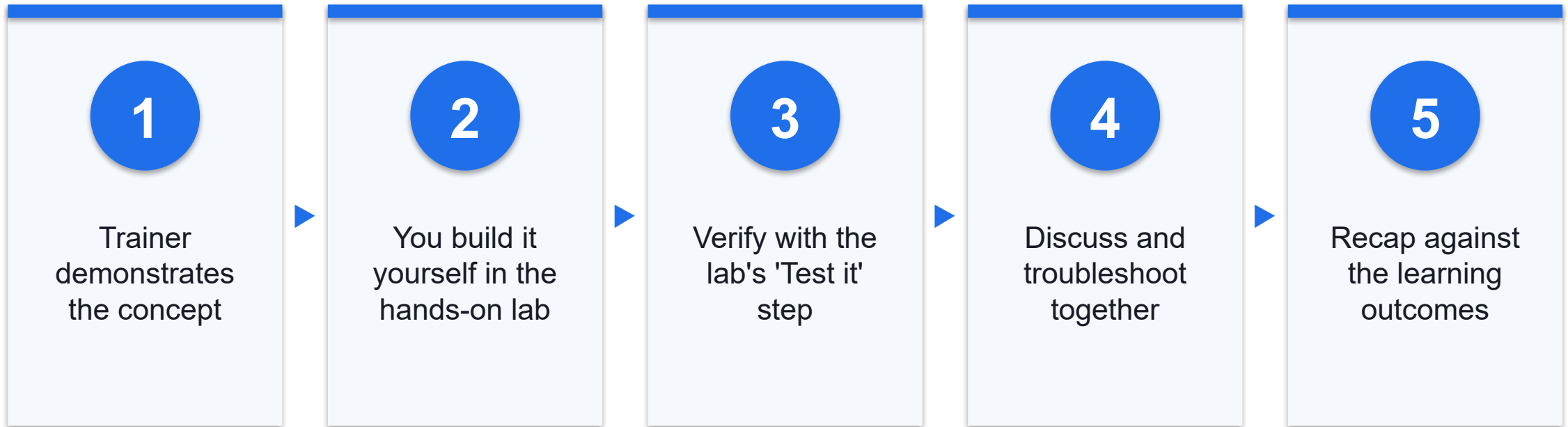
Explain avatar-video architecture and build an n8n workflow that turns a governed brief into a talking-head video through an asynchronous video API.

4

Governed Automation

Automate a monitored video-production and distribution pipeline with approval, retry, accessibility, privacy, cost and human-handoff controls.

How You'll Learn





CORE CONCEPTS

Voice and Video Agent Fundamentals

Four Ideas That Anchor the Course

1

Agent

A bounded workflow that interprets an objective, chooses permitted actions, observes results and verifies completion.

2

Media

Audio and video are typed binary artifacts with format, identity, rights, size and delivery requirements.

3

Orchestration

n8n controls triggers, credentials, branching, retries, asynchronous jobs, evidence and channel adapters.

4

Release

Generation never implies permission to act, impersonate or publish; authority remains explicit.

The Shared Agent Lifecycle

1

Receive

Accept an event, validate its source and assign a request key.

2

Prepare

Normalise audio or brief data and load only approved context.

3

Generate

Transcribe, reason, synthesise speech or render video.

4

Observe

Read real provider state, errors, timestamps and outputs.

5

Verify

Apply safety, factual, media and release checks.

6

Deliver

Return or publish through one authorised channel adapter.

KEY IDEA

Generation is a stage. The workflow owns the outcome.

A reliable media agent proves what it received, what it generated, what it checked and where it delivered the result.

What You Will Build

Voice Core

- Browser audio -> Whisper
- AI Agent -> ElevenLabs
- Binary audio response

Voice Channel

- Telegram voice-note adapter
- Session and fallback handling
- Phone-channel mapping

Video Pipeline

- Governed script and profile
- Create and poll HeyGen job
- Approval and distribution

How Every Lab Progresses

1

Start from the same synthetic HarbourStay guest-experience scenario.

2

Import a credential-free workflow template and reconnect only your own authorised credentials.

3

Inspect the media and data contract before calling a provider.

4

Run a positive path and one bounded failure path.

5

Save a named evidence record so the next lab can reuse the verified checkpoint.

TOPIC 01

Voice Agent

AI voice agents and n8n | webhooks and AI Agent nodes | Whisper and ElevenLabs | browser, chat and phone patterns | service use cases

Key Concepts — Voice Agent

1

A voice agent is a conversational system that accepts speech, derives text and intent, chooses a bounded response, then renders speech; audio alone does not make a workflow agentic.

2

n8n orchestrates triggers, binary audio, model calls, credentials, branches, retries and channel delivery while external models perform transcription, reasoning and speech synthesis.

3

Speech-to-text converts an audio signal into a transcript; the AI Agent interprets that transcript against role, context and policy rather than treating it as an instruction with unlimited authority.

4

Text-to-speech converts approved response text into audio; voice identity, pronunciation, language, pace and output format are production parameters.

5

Browser, messaging and telephone channels expose different audio formats, session identifiers, time limits and response contracts, so the conversation core should be separated from channel adapters.

6

Perceived quality depends on the full latency budget: capture, upload, transcription, reasoning, synthesis and delivery all contribute to the pause a caller experiences.

Key Concepts — Voice Agent (cont.)

7

Conversation state needs an explicit key, bounded memory and expiry; channel identifiers are useful routing data but should not silently become long-term personal profiles.

8

A safe voice agent discloses its automated nature, minimises recordings and transcripts, validates callers before sensitive actions, logs decisions, and hands off when confidence or authority is insufficient.

Voice Agent or Voice Automation?

1

Transcribe

Speech becomes text; no decision is made.

2

Route

A known intent selects a fixed workflow branch.

3

Agent

A bounded model chooses a response or tool.

4

Human

Consequential or uncertain requests leave automation.

The Listen-Think-Talk Loop

1

Listen

Capture -> validate -> transcribe.

2

Think

Interpret -> retrieve -> decide.

3

Talk

Approve text -> synthesise -> deliver.

4

Verify

Trace stages, evidence and fallback.

n8n as the Conversation Orchestrator

1

Trigger

Receive audio and a stable conversation key.

2

Normalise

Produce one transcript request schema.

3

Conversation Core

Apply role, context and response policy.

4

Adapter

Return browser audio, chat audio or call markup.

Audio Transport and Binary Data

1

Browser

Multipart upload from MediaRecorder.

2

Telegram

File ID -> download -> binary property.

3

Phone

Call event plus media stream or recording URL.

4

Provider

Accepted format, limit and response schema.

Whisper Transcription Is Evidence, Not Intent

1

Audio Check

Type, size, duration and signal are usable.

2

Transcribe

Whisper returns text and optional metadata.

3

Normalise

Trim, label language and keep provenance.

4

Clarify

Repeat critical facts instead of guessing.

AI Agent Role and Response Contract

1

Role

Who the agent represents and serves.

2

Scope

Topics and actions it may handle.

3

Response

Answer, clarify or handoff with reason.

4

Evidence

Source, timestamp and uncertainty retained.

ElevenLabs Speech Synthesis

1

Input

Approved response text only.

2

Voice

Authorised voice ID and language fit.

3

Model

Quality, latency and cost choice.

4

Output

Audio format suited to the channel.

Channel Adapters: Browser, Chat and Phone

1

Browser

Webhook in; audio response out.

2

Chat

Message event; file download; reply to chat.

3

Phone

Call event; low-latency turn or callback.

4

Core

Same role, policy and knowledge contract.

Latency Budget and Conversation Turn Design

1

Capture

End-of-speech detection and upload.

2

Listen

Transcription provider time.

3

Think

Context load, tools and model generation.

4

Talk

Synthesis plus channel delivery.

Conversation State, Memory and Idempotency

1

Session

Stable key for related turns.

2

Memory

Minimum context with retention limit.

3

Identity

Separate verification before protected actions.

4

Idempotency

One side effect per event or request.

Failures, Fallbacks and Human Handoff

1

Retry

Bounded backoff for transient provider errors.

2

Clarify

Ask again when speech or intent is unclear.

3

Fallback

Return text or a safe service message.

4

Handoff

Transfer context to a named human queue.

Privacy, Consent and Voice Safety

1

Disclose

Automated identity and recording notice.

2

Minimise

Least audio, transcript and profile data.

3

Authorise

Approved voice, data and tool permissions.

4

Escalate

Human route for consequential requests.

Tool Authority and Transaction Boundaries

1

Inform

Answer from approved service facts.

2

Prepare

Draft an action without committing it.

3

Confirm

Verify identity and explicit intent.

4

Execute

Use one authorised tool and log its result.

Voice Prompt Injection and Untrusted Speech

1

Separate

System policy is not caller-controlled text.

2

Allow-list

Only named tools and actions are reachable.

3

Validate

Check arguments before every side effect.

4

Escalate

Refuse or hand off suspicious requests.

Language, Pronunciation and Inclusive Speech

1

Detect

Observe language; do not guess silently.

2

Confirm

Repeat names, dates and quantities.

3

Pronounce

Test domain terms and abbreviations.

4

Include

Offer text and human alternatives.

Voice-Agent Observability

1

Quality

Grounded answer and clarification rates.

2

Safety

Handoffs, refusals and blocked actions.

3

Latency

Listen, think, talk and delivery time.

4

Cost

Usage per completed useful turn.

Phone Deployment Decision Record

1

Contract

Events, media and response deadline.

2

Identity

Verified factor before protected data.

3

Transfer

Named queue with failure behaviour.

4

Rollback

Disable route and preserve evidence.

Worked Trace: HarbourStay Late Check-In

1

Request

Audio plus request ID; no credentials in payload.

2

Transcript

Observed question retained for traceability.

3

Decision

Information answer; no booking side effect.

4

Delivery

MP3 returned; trace proves each stage.

Hands-On Labs — Voice Agent

Lab 1

- Build a Listen-Think-Talk Web Voice Concierge

Lab 2

- Connect the Concierge to Telegram Voice Notes

Build a Listen-Think-Talk Web Voice Concierge

LAB 1

Build HarbourStay's listen-think-talk core: browser audio to Whisper, a bounded concierge response, ElevenLabs speech and a verified handoff path.

You'll build

A browser voice concierge with one grounded answer, one safe handoff and a redacted execution trace.

Tools: n8n, Whisper, AI Agent, ElevenLabs, browser MediaRecorder

Build a Listen-Think-Talk Web Voice Concierge

STEP 1 OF 2

1

Import the starter, reconnect learner-owned credentials, set the authorised voice ID and run the browser console against the n8n Test URL.

```
$ labs/workflows/lab01-voice-concierge.json  
labs/apps/voice-console/index.html
```

Build a Listen-Think-Talk Web Voice Concierge

STEP 2 OF 2

2

Trace the normal and handoff turns, verify every media stage and save only redacted evidence before activating the Production URL.

\$ Normal: late check-in information
Handoff: change a named booking

Build a Listen-Think-Talk Web Voice Concierge

Test it

Verify a late-check-in question returns approved facts as MP3, while a named-booking change routes to handoff. Preserve the supplied X-C500-Request-ID or use the execution-ID fallback, and record no secrets.

Connect the Concierge to Telegram Voice Notes

LAB 2

Reuse the verified concierge core behind a Telegram voice-note adapter, then specify the extra contract a future phone channel needs.

You'll build

A private voice-note bot with bounded session memory, a no-provider text fallback and a phone-channel map.

Tools: n8n, Telegram, Whisper, AI Agent, window memory, ElevenLabs

Connect the Concierge to Telegram Voice Notes

STEP 1 OF 2

1

Import the Telegram adapter, reconnect credentials and trace voice-file download -> conversation core -> spoken chat reply.

```
$ labs/workflows/lab02-telegram-voice-agent.json
```

Connect the Concierge to Telegram Voice Notes

STEP 2 OF 2

2

Verify voice and fallback paths, then map the extra identity, timing and transfer contract for a phone channel.

```
$ evidence/phone-channel-map.md
```

Connect the Concierge to Telegram Voice Notes

Test it

Verify one spoken reply, a text-only fallback with zero media-provider calls, two connected turns, and a deterministic clean-session test by changing the session epoch.

Recap — Voice Agent

1

You can now: LO1: Explain the listen-think-talk architecture, select bounded voice use cases, and define channel, tool and human-handoff boundaries.

2

You can now: LO2: Build and verify a Whisper-to-AI-Agent-to-ElevenLabs workflow for browser and Telegram voice channels.

TOPIC 02

Video Agent

AI video agents and digital avatars | talking-head video APIs | asynchronous n8n production | marketing, training and service | governed distribution

Key Concepts — Video Agent

1

A video agent turns an objective and approved source material into a planned, generated, checked and distributed media artifact; generation is one stage of the agent workflow.

2

A talking-head request combines script, avatar or talking-photo identity, voice, language, dimensions and optional background or captions into a provider-specific job schema.

3

Video generation is asynchronous: a create request returns a job identifier, while later status checks return processing, completed or failed state and eventually a media URL.

4

The script is the highest-leverage control: it names audience, objective, claim sources, spoken length, call to action and forbidden claims before an avatar renders it persuasively.

5

Consent and provenance must cover the avatar image, voice, music, logos and factual source material; synthetic media should be disclosed when context could mislead viewers.

6

n8n makes the pipeline observable through request IDs, polling limits, error branches, usage logs, approvals and distribution adapters rather than an opaque sequence of API calls.

Key Concepts — Video Agent (cont.)

7

Distribution is a separate decision from generation. A completed file can enter an approval queue, asset library, Telegram channel or social publishing adapter without granting the generator unlimited publishing rights.

8

Video quality includes factual accuracy, pronunciation, captions, visual safety, aspect ratio, accessibility, rendering latency and cost per accepted asset, not just visual realism.

Video Generator or Video Agent?

1

Template

Known text and fixed media slots.

2

Generator

Script and assets become a video.

3

Agent

Plans, renders, observes and routes.

4

Publisher

Separate authority releases the asset.

Talking-Head Video Anatomy

1

Script

Approved spoken words and pronunciation cues.

2

Avatar

Authorised visual identity or talking photo.

3

Voice

Authorised voice ID, language and pace.

4

Canvas

Aspect ratio, background and captions.

HeyGen Asynchronous Job Lifecycle

1

Create

Validate brief and submit one render job.

2

Store

Keep request ID, video ID and submitted settings.

3

Poll

Wait, check state and count attempts.

4

Resolve

Completed URL, failure record or timeout queue.

Script Contract: Brief to Spoken Copy

1

Audience

Who watches and what they already know.

2

Outcome

One useful action after watching.

3

Sources

Approved facts and offer wording.

4

Constraints

Length, tone, claims and disclosure.

Avatar and Voice Consent

1

Owner

Who controls the image, voice and brand assets.

2

Purpose

Permitted messages and audience.

3

Term

Duration, channels and withdrawal process.

4

Disclosure

When viewers must know media is synthetic.

Campaign Profile and Asset Configuration

1

Identity

Avatar and voice IDs with provenance.

2

Format

Dimensions, background, captions and language.

3

Release

Owner, destination and disclosure rule.

4

Version

Effective date and change reason.

Polling, Timeouts and Idempotent Recovery

1

Wait

Respect render time and provider rate limits.

2

Check

Completed, failed or processing state.

3

Limit

Maximum attempts and elapsed deadline.

4

Resume

Continue known job; do not duplicate creation.

End-to-End Production Pipeline

1

Prepare

Brief -> facts -> structured script.

2

Approve

Claims, identity and destination checked.

3

Render

Create once and observe job state.

4

Release

Store, caption, distribute and log.

Distribution Adapters and Release Authority

1

Asset Library

Stable record, provenance and retention.

2

Approval Queue

Named reviewer and release decision.

3

Channel

Caption, media limits and target identity.

4

Rollback

Unpublish, correct and notify owner.

Observability, Cost and Capacity

1

Reliability

Completed renders / accepted requests.

2

Latency

Median and tail time by stage.

3

Quality

Accepted assets / completed renders.

4

Cost

Provider spend / accepted assets.

Accessibility and Release Quality

1

Words

Rendered speech matches approved script.

2

Captions

Accurate, readable and time-aligned.

3

Visual

Safe framing, contrast and correct identity.

4

Alternative

Transcript or text equivalent available.

Video Safety and Content Provenance

1

Source

Approved facts and content owner.

2

Transform

Prompt, script and profile versions.

3

Render

Provider, video ID and check results.

4

Release

Reviewer, destination and published time.

TWO CHECKPOINTS

Moderation Before and After Rendering

1

Brief

Authority, source facts and audience.

2

Script

Claims, length and disclosure approved.

3

Render

Authorised identity and one job ID.

4

Asset

Words, captions, framing and destination.

Scene, Caption and Aspect-Ratio Design

1

Frame

Choose 16:9, 9:16 or 1:1 deliberately.

2

Safe Area

Keep face, logo and text visible.

3

Captions

Readable lines with accurate timing.

4

Profile

Version dimensions and layout settings.

Provider Portability and Stable Contracts

1

Request

Script, profile and idempotency key.

2

Adapter

Translate to provider job schema.

3

Status

Map processing, completed and failed.

4

Evidence

Keep raw job ID and normalised result.

Content Change Control

1

Snapshot

Exact script and release metadata.

2

Hash

Detect any wording change.

3

Version

Bind avatar, voice and layout.

4

Destination

Publish only to reviewed channel.

Rollback and Media Incident Response

1

Stop

Disable schedule and new releases.

2

Contain

Remove or restrict the published asset.

3

Preserve

Keep snapshot, job and decision evidence.

4

Correct

Approve a new version before republishing.

Worked Trace: HarbourStay Welcome Video

1

Brief

Audience, facts, CTA and duration.

2

Approval

Script, identity and destination.

3

Render

One video ID; bounded polling.

4

Release

Checks, channel delivery and provenance.

Hands-On Labs — Video Agent

Lab 3

- Generate a Talking-Head Welcome Video

Lab 4

- Automate a Governed Video Campaign Pipeline

Generate a Talking-Head Welcome Video

LAB 3

Build a two-phase video agent: generate a hashed draft, approve that exact snapshot, create one HeyGen job and poll only its stored video ID.

You'll build

A draft/render webhook, approval-bound browser studio, one captioned video and a status-only resume path.

Tools: n8n, OpenAI Chat Model, HeyGen API, browser Video Studio

Generate a Talking-Head Welcome Video

STEP 1 OF 2

1

Import the two-phase workflow, reconnect credentials and generate one structured draft with an exact SHA-256 approval hash.

```
$ labs/workflows/lab03-avatar-video.json
```

Generate a Talking-Head Welcome Video

STEP 2 OF 2

2

Approve the unchanged snapshot, create one video ID, poll it to a terminal state and use the status-only resume path if the deadline expires.

```
$ labs/workflows/lab03-status-resume.json
```

Generate a Talking-Head Welcome Video

Test it

Prove draft mode makes zero HeyGen calls, render mode rejects a changed script or hash, and one approved snapshot creates one stored video ID that reaches a bounded terminal state.

Automate a Governed Video Campaign Pipeline

LAB 4

Operate one approved video snapshot through persistent queue and run tables, create-once rendering, exact release checks and private distribution.

You'll build

A disabled schedule, persistent idempotency ledger, one private release, duplicate-skip proof and recovery runbook.

Tools: n8n Data Table, HeyGen API, Telegram, Wait, IF, Code

Automate a Governed Video Campaign Pipeline

STEP 1 OF 2

1

Create the exact queue and ledger schemas, import the pipeline and verify approval plus a persistent existing-run query before any provider call.

```
$ labs/workflows/lab04-video-campaign-pipeline.json
```

Automate a Governed Video Campaign Pipeline

STEP 2 OF 2

2

Release once, prove the persisted request key blocks replay, and recover any known timed-out job through the status-only workflow.

```
$ labs/workflows/lab03-status-resume.json
```

Automate a Governed Video Campaign Pipeline

Test it

Verify pending content makes zero external calls, one approved key produces one render and one private post, provider failures write review_required, and replay skips before create.

Recap — Video Agent

1

You can now: LO3: Explain avatar-video architecture and build a governed brief-to-script-to-asynchronous-render workflow.

2

You can now: LO4: Operate an approved, idempotent video pipeline with release checks, private distribution, monitoring and rollback.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Voice Architecture

Explain the listen-think-talk architecture of a voice agent, select suitable business use cases, and design the n8n workflow and channel boundary.

2

Working Voice Agent

Build and verify an n8n voice agent that transcribes audio with Whisper, reasons with an AI Agent, speaks with ElevenLabs, and operates through browser and chat channels.

3

Avatar Video Agent

Explain avatar-video architecture and build an n8n workflow that turns a governed brief into a talking-head video through an asynchronous video API.

4

Governed Automation

Automate a monitored video-production and distribution pipeline with approval, retry, accessibility, privacy, cost and human-handoff controls.

Implementation Roadmap

1

Pilot one informational use case with synthetic or approved low-risk data.

2

Separate the conversation or render core from channel-specific adapters.

3

Add consent, approval, retention, cost and human-handoff controls before broad access.

4

Measure stage latency, errors, accepted outputs and cost before scaling volume.

5

Revalidate provider contracts and permissions whenever a node, model, avatar or destination changes.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your current course material stays available from the C500 course page and GitHub repository.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

C500 | KEEP BUILDING RESPONSIBLY

Thank You!

You can now design, build and verify connected voice and video agent workflows with n8n.